

Test Memo

ECO4128Z

Dr Ayesha Sayed

Senior Lecturer (Finance)

Department of Finance and Tax

University of Cape Town



UNIVERSITY OF CAPE TOWN
IYUNIVESITHI YASEKAPA • UNIVERSITEIT VAN KAAPSTAD

Refer to the figure below, which tracks orange juice futures (ICE-US FCOJ, cents per lb) from March 2023 to March 2025.



Assume today is March 15, 2025, and you're the risk manager at CitrusPro Ltd, a company that supplies frozen concentrated orange juice (FCOJ) to supermarkets across the United States. You are concerned about further declines in the price of orange juice, which could reduce the value of your inventory. You decide to hedge your inventory exposure using futures contracts on the ICE-US. CitrusPro Ltd has 12 million pounds of FCOJ in inventory. The contract size for orange juice futures is 15,000 pounds per contract. The previous trading day's settlement price for the May 2025 delivery contract was 215 cents/lb. The exchange has a daily price limit of 5% up or down relative to the previous day's settlement price. Suppose that a new round of tariffs on citrus exports imposed by President Trump sparks fears of oversupply and causes a massive intra-day decline in prices. The price falls to the daily price limit and the limit remains in effect for the rest of the trading day.

Futures contracts on ICE-US are marked-to-market daily. Assume the following:

- The initial margin requirement is 10% of the contract value.
- The maintenance margin requirement is 7% of the contract value.

1.1 CitrusPro Ltd is concerned that falling orange juice prices will reduce the value of its current inventory. Should the firm take a long or short position in orange juice futures to manage this risk? Based on this position and its underlying business exposure, would CitrusPro Ltd be classified as a hedger or a speculator?

(2 marks)

1.2 Calculate the balance in CitrusPro Ltd's margin account at the end of the trading day on 15 March 2025. Be sure to show all workings, including the daily marking-to-market

(8 marks)



1.3 Determine whether a margin call would be triggered on CitrusPro Ltd's margin account at the end of the trading day on 15 March 2025. Suppose a margin call is required, explain what action the company must take to maintain its hedging position.

(5 marks)



1.4 CitrusPro Ltd is considering entering into a long-term forward contract to supply frozen concentrated orange juice (FCOJ) to a major supermarket chain over the next three years at a fixed price. To hedge against rising prices, the firm is considering using a stack-and-roll strategy, similar to that used by Metallgesellschaft (MG) in 1993–1994. They plan to hedge their long-term commitments using a series of short-dated futures contracts on orange juice, which they would roll over monthly as each contract nears maturity. The firm notes that the orange juice futures market is currently in backwardation, but analysts warn that the market may shift into contango due to increased global supply and new citrus-producing regions entering the market.

Explain what is meant by a stack-and-roll hedging strategy, and describe how it was used by MG to hedge long-term delivery commitments. Further, assume the orange juice market shifts from backwardation to contango. Explain how this would affect CitrusPro's stack-and-roll hedge in terms of roll yield.

(5 marks)

- Definition of stack and roll hedge (1 mark)

- Use by MG (2 marks)

- Impact of backwardation-to-contango and roll yield (2 marks)

QUESTION 2 [20 MARKS]

Nova Bank is a global financial institution based in Luxembourg. It holds significant exposure to the sovereign debt of Argonia, a fictional emerging market country. Concerned about rising political instability and populist rhetoric coming from Argonia's new government, Nova Bank decides to transfer credit risk off its balance sheet without selling the sovereign bonds outright. To do this, the bank enters into credit default swaps (CDSs) referencing Argonia's 10-year euro-denominated sovereign bonds. Nova then packages these CDSs into a synthetic balance sheet collateralized debt obligation (CDO), which is structured into three tranches and sold to a group of institutional investors.

2.1 Explain how a synthetic CDO differs from a cash-flow CDO?

(4 marks)

Cash-funded CDO

Synthetic CDO



4.4 Briefly define overcollateralization in the context of credit enhancement, and determine the overcollateralization of each of the three tranches.

(5 marks)

Definition (2 marks):

[Redacted answer for Definition]

Equity tranche (1 mark):

[Redacted answer for Equity tranche]

Senior and Mezzanine tranches (1 mark):

[Redacted answer for Senior and Mezzanine tranches]

4.5 Alpha Capital has now fully transferred its renewable energy loan exposure into the CDO structured and managed by Vanguard Structured Solutions. After issuance, the following concerns are raised:

- Vanguard pressured the credit rating agency to assign a larger AAA-rated senior tranche than its internal model supported, in order to broaden market appeal and maximise fees.
- The credit rating agency complied and rated 75% of the CDO as AAA, even though its internal model recommended only 60% be rated AAA.
- It is later revealed that two senior analysts at the credit rating agency were hired by Alpha Capital within six months of issuing the rating.

Required:

4.5.1 Define moral hazard and explain how it may arise for:

- Vanguard Structured Solutions as the CDO structurer and equity tranche holder,
- The credit rating agency involved in rating the tranches, and
- Alpha Capital as both the seller of the loans and a subsequent employer of credit analysts.

(5 marks)

Definition (2 marks):

Investment bank
Company that provides financial services, including underwriting, advisory, and asset management, to its clients.

Vanguard Structured Solutions (1 mark):

Vanguard Structured Solutions is a division of Vanguard that provides structured investment solutions, including structured notes and structured funds, to institutional investors.

Credit Rating Agency (1 mark):

Credit Rating Agency is an organization that assigns credit ratings to debt issuers, indicating the creditworthiness of the issuer and the risk of default.

Alpha Capital (1 mark):

Alpha Capital is a private equity firm that provides capital to companies, typically through leveraged buyouts (LBOs) and other private equity transactions.

4.5.2 Use Griffin and Tang (2012) to explain how positive adjustments in CDO ratings create risk for investors.

(2 marks)



4.5.3 Use Cornaggia et al. (2016) to discuss how revolving door dynamics between credit analysts and Alpha Capital can undermine rating integrity and contribute to systemic risk.

(2 marks)

